



J. Kent Dittmer
Vice President, Engineering
P.O. Box 968, Mail Drop PE01
Richland, WA 99352-0968
Ph. 509-377-4348 F. 509-377-2354
jkdittmer@energy-northwest.com

November 25, 2019

GO2-19-158

10 CFR 50.73

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Subject: **COLUMBIA GENERATING STATION, DOCKET NO. 50-397**
LICENSEE EVENT REPORT NO. 2019-001-00

Dear Sir or Madam:

Transmitted herewith is Licensee Event Report No. 2019-001-00 for Columbia Generating Station. This report is submitted pursuant to 10 CFR 50.73(a)(2)(v)(D).

There are no commitments being made to the Nuclear Regulatory Commission herein. If you have any questions, or require additional information, please contact Mr. D.W. Gregoire, Regulatory Affairs Manager, at (509) 377-8616.

Executed on this 25th day of NOVEMBER, 2019.

Respectfully,

KEVIN M. VANSPEYBROECK (DELEGATED FOR KENT DITTMER)
V.P. ENGINEERING

J. Kent Dittmer
Vice President, Engineering

Enclosure: Licensee Event Report 2019-001-00

cc: NRC RIV Regional Administrator
NRC NRR Project Manager
NRC Senior Resident Inspector/988C
CD Sonoda – BPA/1399

**LICENSEE EVENT REPORT (LER)**

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollect.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. Facility Name Columbia Generating Station	2. Docket Number 05000 397	3. Page 1 OF 3
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4. Title LOSS OF DIESEL STARTING AIR DURING MAINTENANCE LEADS TO LOSS OF HIGH PRESSURE CORE SPRAY SYSTEM
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5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
09	24	2019	2019	- 001	- 00	11	25	2019	Facility Name	Docket Number 05000

9. Operating Mode	11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. Power Level	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
100	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ Other (Specify in Abstract below or in NRC Form 366A)	

12. Licensee Contact for this LER	
Licensee Contact Daniel Cook, Principal Engineer, Compliance	Telephone Number (Include Area Code) (509) 377-4745

13. Complete One Line for each Component Failure Described in this Report									
Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
14. Supplemental Report Expected					15. Expected Submission Date				
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No									

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On September 24, 2019, at 2338 PDT during maintenance to drain an air receiver in the High Pressure Core Spray (HPCS) Diesel Starting Air (DSA) system a pressure control valve (PCV) vented, due to a leaking isolation valve, causing the in-service air receiver pressure to drop below the minimum Technical Specifications value. Operations declared the HPCS Diesel Generator, HPCS Service Water, and the HPCS system inoperable.

The root cause of the event was that the plant procedure was revised in 2009 to open the air start motor cross connect valve to retain all 4 air start motors available for diesel start which is not in accordance with current licensing/design bases. Corrective actions include revision to station procedures and repair of the leaking isolation valve.

This is reportable under 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. This event was reported under 10 CFR 50.72(b)(3)(v)(D) as Event Notification #54292.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
		YEAR	SEQUENTIAL NUMBER	REV NO.
Columbia Generating Station	05000- 397	2019	- 001	- 00

NARRATIVE**Plant Conditions**

Before the event, the plant was operating in Mode 1 at 100% power. All Emergency Core Cooling Systems (ECCS) were operable.

Event Description

On September 24, 2019, during maintenance, one of the High Pressure Core Spray (HPCS)[BG] Diesel Starting Air (DSA) [LC] air receivers [RCV] vented from the in-service air header through a DSA Pressure Control Valve (PCV) [PCV] causing the in-service air receiver's pressure to lower.

As part of the maintenance evolution, the DSA air receiver to be drained was isolated from the starting air header and the air start motor cross connect valve [V] was opened per procedure. The isolation valve [V] leaked by creating a lower pressure upstream of the PCV then downstream of the PCV which caused the PCV to relieve the pressure, dropping the in-service air receiver pressure below the Technical Specification operability limit for the HPCS DSA system.

At 2338 PDT operators noted that pressure degraded below the operability limit for the HPCS DSA system. This resulted in the HPCS Diesel Generator (DG)[EK], HPCS Service Water (SW)[BI] and HPCS system being declared inoperable, per plant Technical Specifications.

The leak was isolated and the in-service air receiver pressure was restored at 0104 PDT on September 25, 2019. All Technical Specifications entered due to the degraded air receiver pressure were exited. Total inoperable time was one hour and twenty-six minutes.

This inoperability of HPCS is reportable under 10 CFR 50.73(a)(2)(v)(D) as an event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. An ENS report was made to the NRC at 0526 hours PDT (EN# 54292) on September 25, 2019 pursuant to 10 CFR 50.72(b)(3)(v)(D).

Immediate Corrective Actions

The air start motor cross connect valve was closed and in-service air header pressure was restored resulting in the HPCS DG, HPCS SW, and HPCS system being declared operable.

Assessment of Safety Consequences

Safety was maintained as the Automatic Depressurization System (ADS), Low Pressure Core Spray (LPCS) [BM], and Low Pressure Coolant Injection (LPCI)[BO] mode of Residual Heat Removal (RHR) [BO] system remained operable at the time of the event to mitigate the consequences of an accident. In addition, the Reactor Core Isolation Cooling (RCIC)[BN] system was operable and the HPCS system was maintained available during the period of HPCS inoperability using Dedicated Operators.

Cause of Event

The root cause of the event was that the plant procedure was revised in 2009 to open the air start motor cross connect valve to retain all 4 air start motors available for diesel start which is not in accordance with current licensing/design bases.

The direct cause of the event was leak by of the isolation valve. A contributing cause was that a Condition Report to assess the controlling procedure for the maintenance was not fully evaluated.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Columbia Generating Station	05000-397	YEAR 2019	SEQUENTIAL NUMBER 001	REV NO. 00

NARRATIVE

Similar Events

A similar event happened on July 24, 2017. While attempting to drain only one HPCS DSA air receiver, the PCV started venting the other air receiver at the same rate, maintaining equal pressure between the receivers. HPCS was already inoperable due to maintenance at the time of this previous occurrence.

Corrective Actions

Revise the Standard Operating Procedure for Diesel Starting Air to remove steps to open the air start motor cross connect valve when draining an air receiver.

Repair the leaking isolation valve.

Revise station procedures to provide additional guidance when assessing Condition Reports.

Energy Industry Identification System (EIIS) Information codes from IEEE Standards 805-1984 and 803-1983 are represented in brackets as [XX] and [XXX] throughout the body of the narrative.